

## UPSALA, ON, Canada

WMO: 714350

Lat: 49.0361N

Lon: 90.4614W

Elev: 489

StdP: 95.59

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                   | (o)  | (p)   |
| 1             | -33.5      | -30.4 | -37.2                         | 0.1 | -33.0 | -34.2 | 0.2 | -30.0 | 7.2                   | -12.7 | 6.4 | -13.1 | 1.4                   | 310  | 0.612 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.9                   | 28.1            | 19.7 | 26.3 | 18.6 | 24.8 | 17.5 | 21.4                | 26.2 | 20.1 | 24.4 | 19.0 | 23.0 | 3.5                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.7                            | 15.3 | 24.2 | 18.6 | 14.2 | 22.6 | 17.5 | 13.3 | 21.2 | 64.8          | 26.3 | 60.0 | 24.5 | 56.2 | 23.0 | 25.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 6.7           | 5.8  | 5.2 | DB                         | -37.7 | 30.6               | 3.6 | 1.3                                                | -40.3 | 31.5       | -42.4 | 32.3       | -44.4 | 33.0       | -47.1 | 34.0 | (4) |
| (5)               |      |     | WB                         | -37.6 | 22.8               | 3.3 | 1.2                                                | -40.0 | 23.6       | -41.9 | 24.3       | -43.7 | 25.0       | -46.1 | 25.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 1.8    | -16.4 | -13.8 | -6.7  | 1.4   | 9.0   | 14.8  | 17.3  | 16.0  | 11.8  | 4.0   | -4.2  | -12.0 |  |  |
|                                                                        | DBStd       | 13.06  | 8.04  | 7.54  | 7.45  | 5.53  | 4.78  | 3.88  | 3.34  | 3.50  | 4.71  | 4.81  | 6.33  | 7.53  |  |  |
|                                                                        | HDD10.0     | 3687   | 818   | 666   | 520   | 261   | 77    | 7     | 0     | 1     | 34    | 197   | 425   | 683   |  |  |
|                                                                        | HDD18.3     | 6080   | 1076  | 899   | 778   | 508   | 291   | 117   | 61    | 89    | 201   | 445   | 674   | 941   |  |  |
|                                                                        | CDD10.0     | 712    | 0     | 0     | 1     | 3     | 46    | 151   | 225   | 187   | 88    | 10    | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 63     | 0     | 0     | 0     | 0     | 2     | 11    | 27    | 17    | 5     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 689    | 0     | 0     | 0     | 1     | 46    | 154   | 274   | 174   | 41    | 1     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 124    | 0     | 0     | 0     | 0     | 8     | 28    | 57    | 28    | 4     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.3    | 2.3   | 2.3   | 2.4   | 2.4   | 2.4   | 2.2   | 2.2   | 2.1   | 2.2   | 2.4   | 2.4   | 2.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 807    | 38    | 31    | 39    | 61    | 83    | 105   | 100   | 81    | 100   | 71    | 55    | 45    |  |  |
|                                                                        | PrecMax     | 1000   | 71    | 60    | 78    | 151   | 150   | 197   | 167   | 137   | 199   | 154   | 115   | 83    |  |  |
|                                                                        | PrecMin     | 612    | 12    | 9     | 15    | 17    | 38    | 50    | 64    | 29    | 32    | 29    | 17    | 20    |  |  |
|                                                                        | PrecStd     | 91     | 13    | 13    | 17    | 26    | 27    | 36    | 28    | 28    | 41    | 29    | 25    | 15    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 2.4   | 6.4   | 14.2  | 19.7  | 27.7  | 29.6  | 30.6  | 29.4  | 26.8  | 20.8  | 12.5  | 3.9   |  |  |
|                                                                        |             | MCWB   | 0.5   | 3.1   | 8.8   | 10.2  | 17.0  | 20.7  | 21.3  | 21.2  | 19.8  | 14.9  | 8.2   | 1.7   |  |  |
|                                                                        | 2%          | DB     | 0.0   | 2.2   | 9.0   | 16.4  | 24.2  | 26.7  | 28.2  | 27.1  | 24.2  | 17.5  | 9.1   | 1.2   |  |  |
|                                                                        |             | MCWB   | -0.6  | 0.4   | 4.7   | 8.3   | 14.7  | 18.1  | 20.0  | 19.7  | 18.0  | 11.9  | 6.3   | 0.2   |  |  |
|                                                                        | 5%          | DB     | -1.8  | 0.2   | 6.0   | 13.6  | 21.5  | 25.0  | 26.3  | 25.2  | 22.1  | 14.6  | 6.1   | 0.0   |  |  |
|                                                                        |             | MCWB   | -2.5  | -1.1  | 2.2   | 6.8   | 12.9  | 16.5  | 18.9  | 18.7  | 16.9  | 10.9  | 4.0   | -0.5  |  |  |
|                                                                        | 10%         | DB     | -4.4  | -2.2  | 3.2   | 11.0  | 18.7  | 23.2  | 24.5  | 23.5  | 20.0  | 11.8  | 3.9   | -1.5  |  |  |
|                                                                        |             | MCWB   | -5.0  | -3.4  | 0.3   | 5.1   | 11.5  | 15.6  | 17.8  | 17.8  | 15.9  | 8.9   | 2.5   | -2.2  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 0.9   | 3.3   | 10.7  | 12.6  | 18.9  | 22.2  | 22.8  | 22.7  | 21.1  | 16.0  | 9.9   | 2.1   |  |  |
|                                                                        |             | MCDB   | 1.7   | 5.2   | 12.0  | 16.8  | 24.3  | 28.2  | 28.2  | 26.9  | 25.3  | 18.7  | 11.5  | 3.4   |  |  |
|                                                                        | 2%          | WB     | -0.6  | 0.6   | 5.2   | 9.5   | 16.5  | 19.9  | 21.5  | 21.1  | 19.4  | 13.3  | 6.6   | 0.5   |  |  |
|                                                                        |             | MCDB   | 0.0   | 1.8   | 8.3   | 14.8  | 21.5  | 24.6  | 26.3  | 25.8  | 22.7  | 16.4  | 8.8   | 1.1   |  |  |
|                                                                        | 5%          | WB     | -2.5  | -1.1  | 2.5   | 7.3   | 14.2  | 18.2  | 20.1  | 19.7  | 17.7  | 11.2  | 4.3   | -0.6  |  |  |
|                                                                        |             | MCDB   | -1.8  | 0.4   | 5.3   | 12.9  | 19.3  | 22.4  | 24.4  | 23.8  | 20.7  | 14.2  | 5.8   | 0.1   |  |  |
|                                                                        | 10%         | WB     | -5.0  | -3.4  | 0.6   | 5.5   | 12.3  | 16.9  | 19.0  | 18.5  | 16.2  | 9.0   | 2.5   | -2.1  |  |  |
|                                                                        |             | MCDB   | -4.4  | -2.2  | 2.7   | 10.5  | 17.6  | 21.2  | 23.0  | 22.3  | 19.4  | 11.4  | 3.9   | -1.6  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 10.6  | 12.2  | 12.8  | 12.1  | 13.0  | 12.4  | 11.9  | 12.0  | 11.0  | 8.6   | 7.4   | 8.9   |  |  |
|                                                                        | 5% DB       | MCDBR  | 9.3   | 11.6  | 13.1  | 16.6  | 17.5  | 15.6  | 14.0  | 14.0  | 13.8  | 13.9  | 9.6   | 8.6   |  |  |
|                                                                        |             | MCWBR  | 8.6   | 9.6   | 9.0   | 9.9   | 9.2   | 7.9   | 7.2   | 7.8   | 8.3   | 9.2   | 7.0   | 7.8   |  |  |
|                                                                        | 5% WB       | MCDBR  | 8.9   | 10.4  | 11.6  | 15.0  | 14.1  | 12.6  | 12.0  | 12.4  | 11.6  | 12.3  | 8.7   | 8.4   |  |  |
|                                                                        |             | MCWBR  | 8.3   | 8.7   | 8.5   | 9.7   | 8.3   | 7.3   | 7.0   | 7.6   | 7.6   | 8.8   | 6.8   | 7.7   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.248 | 0.256 | 0.286 | 0.327 | 0.342 | 0.365 | 0.377 | 0.360 | 0.346 | 0.301 | 0.275 | 0.251 |  |  |
|                                                                        | taud        |        | 2.368 | 2.347 | 2.377 | 2.422 | 2.451 | 2.403 | 2.397 | 2.462 | 2.486 | 2.589 | 2.543 | 2.416 |  |  |
|                                                                        | Ebn at Noon |        | 833   | 911   | 935   | 923   | 918   | 894   | 877   | 879   | 858   | 853   | 803   | 783   |  |  |
|                                                                        | Edh at Noon |        | 65    | 84    | 98    | 105   | 107   | 113   | 112   | 101   | 89    | 68    | 56    | 54    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.21  | 2.40  | 3.68  | 4.82  | 5.27  | 5.58  | 5.72  | 4.88  | 3.52  | 2.05  | 1.08  | 0.84  |  |  |
|                                                                        | RadStd      |        | 0.17  | 0.25  | 0.31  | 0.47  | 0.30  | 0.44  | 0.36  | 0.40  | 0.28  | 0.28  | 0.12  | 0.10  |  |  |

## Historical Trends

|                              |  | DBAvg |     | Heating |        | Cooling |       | Degree-Days |         |         |         |
|------------------------------|--|-------|-----|---------|--------|---------|-------|-------------|---------|---------|---------|
|                              |  |       |     | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP       | HDD10.0 | HDD18.3 | CDD18.3 |
|                              |  | (d)   | (e) | (f)     | (g)    | (h)     | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A | N/A     | N/A    | N/A     | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (15 neighbors) |  | N/A   | N/A | N/A     | N/A    | N/A     | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page